

Sharing between teams — how a squad works when the boats have different owners

The design question: an Olympic squad trains together, but the boats belong to different people — different crews, different campaigns, sometimes different nations — each with its own data, its own coach relationship and its own privacy. They need to see each other's tracks. How?

First, what a TEAM is, because getting this wrong distorts everything after it. A team in SSA is a *programme or organisation*, not a boat. It routinely holds several boats: a campaign that kept Northstar 72 when Northstar 76 arrived; a national squad running two 49ers under one management. Those boats share a team because they share an owner — and they need no squad at all, since one team's boats are already visible to that team. Splitting them into separate teams to make them "peers" would be cargo-culting the squad model onto data that was never divided.

A squad is for the other case: **boats under different ownership**, where visibility has to be negotiated rather than assumed. The two cases coexist in one venue all the time — three teams in a squad, one of which fields two boats — and the schema below handles that without a special case, because membership is keyed on the team and sharing is decided per session.

The short answer: a SQUAD that teams join by consent, sharing is scoped to what a squad is actually for (days trained together), and reciprocity is enforceable. Reads are squad-wide by default; WRITES stay with the owning team unless it says otherwise.

1. Why this is not the problem the other tools solved

Every comparable tool assumes **one owner**. Njord's sharing is "invite teammates to the boat so they can access all data" — per user, whole boat, no scope, no expiry. Kinetix, SailViewer, Vantage and SailSync are all single-coach: the coach holds everything and shows it in a joint debrief.

That model breaks the moment the boats are peers rather than possessions:

- A 49er crew's data belongs to **that crew**, not to whoever is coaching this week. Coaches change; the data should not change hands.
- Two boats in the same national squad are **selection rivals**. They will share a training block and absolutely not a season.
- The person who collects the files after sailing is often a coach who works with several of the boats but owns none of them.

So the primitive is not "invite a user to my boat". It is **"our two teams agree to train together, on these terms, for this period"**.

The one good precedent: Hudl

Not sailing — American high-school sport, where rival schools exchange game film. Three modes worth stealing from:

Hudl	what it is	worth taking?
Direct exchange	team admin picks another school, selects up to 5 videos	yes — the handshake and the explicit, finite selection
Focus Exchange Network	video shared automatically when two schedules match	the idea: sharing keyed on a SHARED EVENT, not on a standing grant
League pool	contribute to the pool and you get access to what others contributed — <i>you cannot download a game unless you have uploaded one</i>	yes. Enforced reciprocity is exactly right for rivals

That last one is the best idea in the space and nobody in sailing has it.

What SSA already has, and should not reinvent

- `video_shares` — a capability token with `expires_at`, `revoked_at`, `include_overlay` (*what* is shared, not just whether), `view_count` and `last_viewed_at`. Time-bounded, revocable, content-scoped, audited. The instincts here are already right; this design reuses them.
- `invitations` — a handshake with a role, an optional boat, a validity window, `max_uses`, `auto_approve`, expiry and revocation.
- `subteams` — grouping *within* a team. Not this problem, and worth saying so plainly: a squad is not a subteam.

2. The shape

squads	a named training relationship
	"NED 49er squad", "Palma winter block"
├─ squad_members	one row per TEAM that joined, with its terms
	├─ team_id who joined
	├─ boat_id which boat they are contributing
	├─ status invited active left
	├─ shares[] what they contribute (see §3)
	├─ allow_coach_upload can squad coaches upload FOR this boat?
	└─ valid_from / to the block, not for ever
└─ training_days.squad_id	the day several boats sailed together

A team joining a squad is a **negotiated act by that team's manager or coach**, recorded with its own terms, revocable, and time-bounded. It is not a permission an admin hands out.

Why the day, not the boat, is the unit of sharing

The obvious design is "Team A shares boat X with Team B". It is also the wrong one, for a reason that only shows up in this sport: **the teams are rivals**. A standing boat-level grant means "you can watch everything I ever do", which no selection rival will agree to, so the feature goes unused and everyone goes back to WhatsApp.

What they *will* agree to is "we sailed together on these days, let's compare those days". That is bounded, obviously fair, and exactly what happened.

So: `training_days` gains a `squad_id`. A session is visible across the squad **when its training day is a squad day and both teams were active members that day**. Leave the squad and future days stop being shared; the days you were there for stay, because they were shared at the time and pretending otherwise would be a lie about what other people already have.

3. What "shared" means — per category, not one switch

A track is a fact about where a boat sailed. A debrief recording is a crew talking about their own mistakes. Treating those the same is the mistake that makes teams refuse to share anything at all.

category	default	why
track + derived analysis (positions, speed, TWA, phases)	on	this is the thing a squad is for
manoeuvre and speed-test results	on	the comparison itself
photos	off	usually fine, occasionally not — let them choose
video	off	crew audio. Share a clip deliberately (<code>video_shares</code> already does this well)

category	default	why
debriefs, notes, tags	off, and not offerable at first	a team's own words about its own performance
rig settings / tuning numbers	off	the crown jewels. Some squads share, most do not

Per-category, per-member, stored on `squad_members.shares`. The default is the narrow one; opening up is a decision each team makes for itself.

Reciprocity

`squads.reciprocal` (default **true**), the Hudl pool rule: **you see a day's other boats only if you contributed your own track for that day**. It is self-enforcing, it needs no arbitration, and it answers the objection every rival raises first — "what stops them taking and never giving?"

4. Writes: the delicate half

Reads are the easy case. The request — "each team can both upload and view other boats' logs, including multiple-boat logfile drops" — needs one person to write into several teams' boats. That is a much stronger grant than reading, and it should not ride in on the back of a read agreement.

Two routes, and the first is already built:

1. **The coach holds a membership in each team.** Memberships are per team and already support this; a coach working with three boats has three memberships. Nothing new is needed, and the audit trail is honest: the coach uploaded, as themselves, to a team that admitted them.
2. `squad_members.allow_coach_upload` — a team opts in to letting the squad's coaches upload on its behalf. Off by default. This is the convenience the multi-file drop wants, and it stays a decision of the team being written to.

Either way the file is routed to the **owning** boat and the session belongs to the owning team. A squad never owns data; it is an agreement about visibility, not a container. That single rule keeps leaving a squad clean: nothing has to be untangled or handed back.

`tracker-import` then grows a per-file `--team`, and the coach's own memberships (or the flag) decide whether each write is allowed. The route it uses already runs as the user, so RLS is the authority — which is the property to preserve.

5. What this looks like on the water

The morning brief: five 49ers and a RIB, under four different campaigns — one of them fields two of the boats. So: four teams, one holding two boats, all four in the squad. The squad coach is a member of all four, or three, or none.

That mixed shape is the normal one, and nothing in the design treats it as a special case: the two-boat team already sees both its own boats, and the squad decides the other three.

- **After sailing**, whoever has the files runs one import. Each file is routed to its own boat, in its own team. Boats that did not share stay private and their file simply is not there.
- **Wind is pooled** across everyone who contributed that day — which is the accuracy argument, not a nicety: 3–5° from one boat, ~1° from five.
- **Each crew opens their own session** and sees the squad's other tracks on the map, toggleable, exactly as the current multi-boat map already draws them. The two-boat team sees its stablemate by the same mechanism, without having shared anything. They see tracks and speed; they do not see anyone's debrief.
- **A crew that left the squad in March** still has the January days. Their April days were never shared.

6. Schema sketch

```
CREATE TABLE squads (  
  id UUID PRIMARY KEY, name TEXT NOT NULL, note TEXT,  
  reciprocal BOOLEAN NOT NULL DEFAULT TRUE,      -- the Hudl pool rule  
  created_by_user_id UUID, created_at TIMESTAMPTZ  
);  
  
CREATE TABLE squad_members (  
  id UUID PRIMARY KEY,  
  squad_id UUID NOT NULL REFERENCES squads(id) ON DELETE CASCADE,  
  team_id UUID NOT NULL REFERENCES teams(id) ON DELETE CASCADE,  
  -- DROPPED before shipping: see §8. A team joins ONCE, with all its boats,  
  -- and decides per SESSION what it shares. 0071 has no boat_id here.  
  boat_id UUID NOT NULL REFERENCES boats(id) ON DELETE CASCADE,  
  status TEXT NOT NULL CHECK (status IN ('invited','active','left')),  
  -- what this team contributes. Narrow by default; each team decides.  
  shares JSONB NOT NULL DEFAULT '{"track":true,"manoeuvres":true,  
    "photos":false,"video":false,"debriefs":false,"rig":false}',  
  allow_coach_upload BOOLEAN NOT NULL DEFAULT FALSE,  
  valid_from DATE, valid_to DATE,  
  invited_by_user_id UUID, decided_by_user_id UUID, decided_at TIMESTAMPTZ,  
  UNIQUE (squad_id, team_id, boat_id)  
);  
  
ALTER TABLE training_days ADD COLUMN squad_id UUID REFERENCES squads(id);
```

Visibility, as one predicate — a session is readable across a squad when:

```
its training_day.squad_id = S  
AND the reader's team is an ACTIVE member of S, within its validity window  
AND the owning team is an ACTIVE member of S, within its window, on that date  
AND the owning member's shares allow the category being read  
AND (NOT S.reciprocal OR the reader's own boat also contributed that day)
```

That is one RLS helper, `has_squad_access(session_id, category)`, alongside the existing `has_boat_access`. Everything else in the app keeps working unchanged because it already asks the same question through RLS.

7. What NOT to build

- **A squad that owns boats or sessions.** It is an agreement about visibility, nothing more. The moment a squad owns data, leaving one becomes a migration.
- **One global "share everything" switch.** §3 is the whole point.
- **Silent sharing.** Every member sees who else is in the squad, what each contributes, and when the window ends — before joining, not after.
- **Cross-team writes by default.** §4.
- **Pretending revocation is retroactive.** Say plainly that leaving stops future days and that what was shared has been seen.

8. Open questions, worth deciding before building

- **Does a squad need its own coach role**, or is "a user with a membership in several member teams" enough? The second needs no new concept and I lean to it; the first would let a squad hand over coaching without touching five teams' memberships.
 - **Who creates a squad?** Any team manager, or only on invitation from one? The first is simpler and matches how people actually organise.
 - **~~Is `boat_id` on `squad_members` right**, or should a team contribute several boats to one squad?~~ **Decided: no `boat_id`**. As shipped in 0071, `squad_members` is `UNIQUE (squad_id, team_id)` with no boat column, and `shares_squad_with()` reasons purely about teams. A team with two 49ers joins once and shares whichever *sessions* it chooses, boat by boat — which is strictly more expressive than a per-boat membership, and avoids the question of what happens when a team acquires a boat mid-block. The sketch in §6 still shows the `boat_id` column; the migration is the authority, not the sketch.
 - **Regattas**. A squad day is training. Does the same machinery cover "we all sailed the same regatta, pool it"? Probably — a regatta is a day with a shared venue and an event tracker — but the tracker's own roster may make squad consent redundant there.
 - **Does reciprocity count a day where a boat sailed but its file is late?** A grace period is kinder than a hard rule, and a hard rule is easier to explain. I would start hard and soften it if it bites.
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Sources

- [Hudl — exchange video with a Hudl team](#) · [League pool \(contribute to access\)](#) · [Focus Exchange Network](#)
- [Njord user guide — overview, incl. Sharing Boat Access](#)
- SSA's own precedents: `video_shares` (0053), `invitations`, `subteams` (0031), and `training_days` / `trackers` (0070).